FEDERAL SPECIFICATION

WHEELS, LATHE, SILICON CARBIDE, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers silicon carbide wheels for attachment to standard dental stone lathe chucks suitable for dental laboratory use.

1.2 Classification.

1.2.1 *Type, size and classes.* — Wheels, lathe, silicon carbide, covered by this specification shall be of one type, one size, and of the following classes, as specified in the invitation for bids:

Class A.—Medium grit
Class B.—Coarse grit

2. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

2.1 Specifications. — There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Other publication.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

National Bureau of Standards Simplified Practice Recommendation R118–45, Abrasive Grain Sizes.

(Copies of the National Bureau of Standards Simplified Practice Recommendation R118–45, Abrasive Grain Sizes, may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Abrasive.* — The abrasive shall be high quality silicon carbide crystals, free from extraneous material and suitable for the purpose intended.

3.1.2 *Binder.*—The binder shall be a fused ceramic.

3.1.3 *Hub.* — The hub shall be made of metal suitable for the purpose intended.

3.2 Construction. — The wheels shall be true in form with square edges, so constructed that they will wear true and evenly.

3.2.1 *Hub.*—Wheels shall be provided with a metal hub which shall be drilled with a hole ¼ inch in diameter for attachment to a standard ¼ inch stone lathe chuck. The metal center when drilled, shall serve as a bushing. The hub shall be flush with the ceramic part of the wheel on both sides.

3.3 Fineness.

3.3.1 *Class A.*—The size of the abrasive grain for class A wheels shall be grit No. 180

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in conformance with table I of Simplified Practice Recommendation R118–45.

3.3.2 *Class B.*—The size of the abrasive grain for class B wheels shall be grit No. 100 in conformance with table I of Simplified Practice Recommendation R118–45.

3.4 **Sizes.**—Wheels of both classes shall be approximately 2 inches in diameter and ¼ inch in thickness.

3.5 **Workmanship.**—Workmanship shall be first class in every respect. The wheels shall be symmetrically made with chuck holes truly centered and shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Wheels.*—Unless otherwise specified, one wheel from each class shall be taken at random from each delivery of 100 or fraction thereof.

4.1.2 *Abrasive.* — Unless otherwise specified, at least 400 grams of each abrasive grain size shall be furnished the laboratory for testing purposes to determine compliance with specifications. Contractor shall certify that the abrasive furnished represents the material employed in the fabrication of the wheels delivered.

4.2 **Inspection.**—Wheels shall be inspected for material, classes, size, workmanship, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Construction.*—To determine conformance to the requirements of 3.2, three wheels of each class taken at random shall be mounted on a standard dental stone lathe chuck and rotated against any hard substance.

4.3.2 *Fineness.*—To determine conformance to the requirements of 3.3, a 100 gm. sample of each abrasive grain size shall be shaken for 5 minutes on a Rotap or equivalent machine operated at a speed of 290 revolutions per minute and 156 taps per minute. Where more than a single sieve is used, they shall be agitated simultaneously. U. S. Standard sieves referenced in Simplified Practice Recommendation R118–45 shall be used.

5. PREPARATION FOR DELIVERY

5.1 **Packaging.**—Unless otherwise specified, wheels shall be packaged in accordance with accepted commercial practice.

5.2 **Packing.**—The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, packages shall be marked with the name, class, and size of the commodity, the quantity contained therein, and the names of the manufacturer and contractor.

5.3.2 *Shipping containers.*—Unless otherwise specified, shipping containers shall be marked with the name, class, and size of the commodity as defined by the contract or order under which the shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any desired options offered herein. (See 1.2, 4.1, 5.1, 5.3, 6.2 and 7.2.6.)

6.2 It is believed that this specification adequately describes the characteristics nec-

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essary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification covers the following items appearing in the Armed Services Catalog of Medical Matériel:

5-689-000—Wheel, Silicon Carbide, Square Edge, 2 Inch Diameter, ¼ Inch Thick, Coarse.

5-690-000—Wheel, Silicon Carbide, Square Edge, 2 Inch Diameter, ¼ Inch Thick, Medium.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally procured for Federal Government requirements.

6.5 Transportation description.

6.5.1 *Transportation description applicable to these items is:*

Wheels, abrasive, other than pulp grinding.

In barrels, boxes or crates, also carload or truckload, loose, packed in packing material.
Carload weight 30,000 pounds.
Truckload weight 21,400 pounds.

Patent notice.—When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN-B-601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621—Boxes; Wood, Nailed and Lock-Corner.

NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

MIL-B-107—Boxes, Wood, Wirebound (Overseas Type).

JAN-P-120—Packaging and Packing for Overseas Shipment—Cartons, Folding, Paperboard.

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN-P-133—Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

MIL-A-140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

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Navy Department Specification:

General Specifications for Inspection of Material. [1]

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Packaging.*

7.2.2.1 *Unit packaging.*—One wheel, of the class specified, constitutes one unit. Each unit shall be packaged in a carton of appropriate size, conforming to the requirements of Military Specification JAN-P-120 or JAN-P-133. Each carton shall be adequately secured to prevent accidental opening.

7.2.2.2 *Intermediate packaging.* — Twelve unit cartons (12 units) either shall be packaged in a carton of appropriate size and design, or be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

7.2.3 *Packing.*

7.2.3.1 *Domestic.*—Twenty-four intermediate packages (288 units) shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, NN-B-621, or NN-B-631.

7.2.3.2 *Overseas.*—Twenty-four intermediate packages (288 units) shall be packed in a shipping container designed for type 2 load, and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall conform to the requirements of the appendix to the applicable box specification. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN-P-125 for case liners; seams and closures shall be completely sealed by a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification MIL-A-140.

7.2.3.3 *Packing variation.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.4 *Marking.*—Each unit carton, each intermediate package and each shipping container shall be marked in accordance with Military Standard MIL-STD-129.

7.2.5 *Inspection procedures (Navy only).* —The general inspection procedures shall be in accordance with Department of the Navy General Specifications for Inspection of Material.

7.2.6 *Ordering data.*—Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.3.)

NAVY INTEREST: Md

[1] Applicable to Navy purchases only.

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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